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EDUCATION

South-Central Minzu University Wuhan, China
M.S. in Computer Technology *Sep 2024 – Jun 2027 (expected)* | *GPA: 3.51/4.0*
South-Central Minzu University Wuhan, China
B.E. in Network Engineering *Sep 2020 – Jun 2024* | *GPA: 3.17/4.0*

RESEARCH INTERESTS

Medical Image Segmentation, Multimodal Learning, Semi-Supervised Learning

PUBLICATIONS

Yang, Y., Hong, Y., Yu, W., et al. HSFPN-Det: An Effective Model for Detecting Rice Pests and Diseases. *The Visual Computer (TVC)*, 2025. [Paper]. (**JCR Q2, IF: 3.4**).

Yang, Y., Zhu, J., Yang, B., et al. An Improved YOLOv8-based Rice Pest and Disease Detection Method. *Computer Graphics International (CGI)*, 2024. [Paper]. (**CCF C**).

Yang, Y., Huang, X., Zhang, X., et al. Learning Symmetric Cross-Modal Representations via Feature Distillation and Contrastive Fusion for Brain Tumor Segmentation. [Under Review].

Yang, Y., Huang, X., Zhang, X., et al. Background as Parts: Sub-Background Modeling for Uncertainty-Aware Semi-Supervised Medical Image Segmentation. [Under Review].

Yang, Y., Huang, X., Zhang, X., et al. Hi-ERMNet: Hierarchical Transformer with Edge Refinement for Retinal Vessel Segmentation. [Under Review].

Huang, X., **Yang, Y.**, Lv, Y., et al. Ontology-Guided Lesion-Aware Vision-Language Alignment for Retinal Image-Text Retrieval. [Under Review].

RESEARCH EXPERIENCE

Semi-Supervised Medical Image Segmentation Jan. 2026 – Apr. 2026
Graduate Researcher, South-Central Minzu University *Wuhan, China*

- ***Project Title:*** Research on Semi-Supervised Medical Image Segmentation Based on Sub-Background Modeling
- ***My Responsibilities:*** Proposed BAP, a semi-supervised medical image segmentation framework that advances conventional unified background modeling into structured sub-background modeling. Achieved Dice scores of 90.55%, 88.64%, and 86.80% on LA, ACDC, and BraTS2019 with 10% labeled data, respectively.

Brain Tumor Segmentation Jul. 2025 – Nov. 2025
Graduate Researcher, South-Central Minzu University *Wuhan, China*

- ***Project Title:*** Research on Brain Tumor Segmentation Based on Bidirectional Feature Distillation and Multimodal Feature Fusion
- ***My Responsibilities:*** Proposed SFDMNet based on feature distillation and contrastive learning, improving over the baseline by 4.30% and 1.62% on public benchmarks. Generalization experiments on the abdominal multi-organ dataset CHAOS further verified its generalization capability.

Retinal Fine Vessel Segmentation Apr. 2025 – Sept. 2025
Graduate Researcher, South-Central Minzu University *Wuhan, China*

- ***Project Title:*** Research on Key Technologies for Multimodal Intelligent Analysis of Retinal Fine Vessels and Disease Prediction
- ***My Responsibilities:*** Applied the Sobel operator for edge detection on retinal images and proposed Hi-ERMNet to segment fine retinal vessels. Simulated non-ideal conditions with Gaussian blur to verify the robustness of Hi-ERMNet.

SELECTED HONORS & AWARDS

- National Scholarship for Masters Students — Sept. 2026
- Outstanding Graduate Student — Sept. 2026
- First-class Academic Scholarship for Graduate Students — Sept. 2026
- First-class Academic Scholarship for Graduate Students — Sept. 2025
- Outstanding Graduate Student — Sept. 2025
- Second-class Academic Scholarship for Graduate Students — Sept. 2024
- National Encouragement Scholarship — Sept. 2023

SKILLS AND SOFTWARE

Programming: Python, C/C++, Java
Tools: Git, Linux
Scientific Writing: LaTeX
Frameworks: PyTorch
Data Analysis & Visualization: Numpy, Scipy, Matplotlib

REVIEW SERVICES

- Expert Systems With Applications
- Applied Soft Computing
- Journal of Computer-Aided Design & Computer Graphics
- The Visual Computer
- ACM MM
- Computer Graphics International

LANGUAGES

Chinese Mandarin (native), English (CET6 504)